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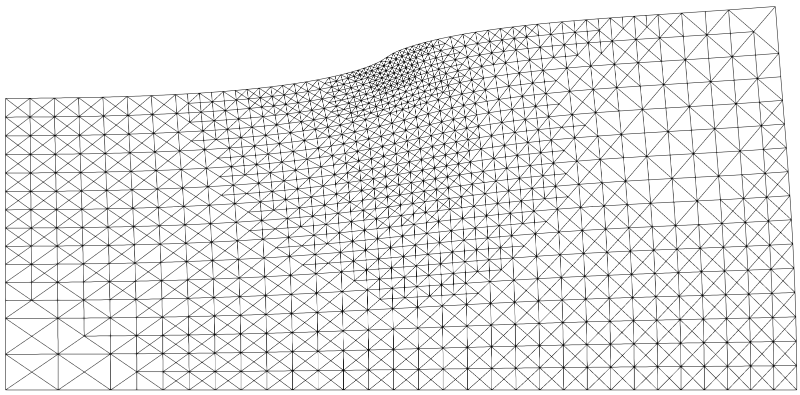

TME250 FEM – Solids

Computer Assignment 3:

Adaptive Finite Element Analysis

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Preface

The aim of Computer Assignment 3 is to construct an adaptive algorithm to refine a mesh to reduce the error in the solution with respect to different error estimators. The finite element toolbox CALFEM may be used.

The task is to be carried out individually and an informal report, containing explicit derivations, written codes and conclusions, is to be handed in the latest on **Friday, January 9, 2026**.

The reports, as well as the code, will automatically be checked for plagiarism. It is okay to discuss the implementation with other students, but DO NOT share code with each other. A well written report, together with correct implementation, can give maximum 2 points. A late hand in, without prior agreement with the examiner, will give at maximum 1 point. Contact us in good time if you need to postpone your hand in!

Göteborg in Dec. 2024

Carl Larsson and Fredrik Larsson

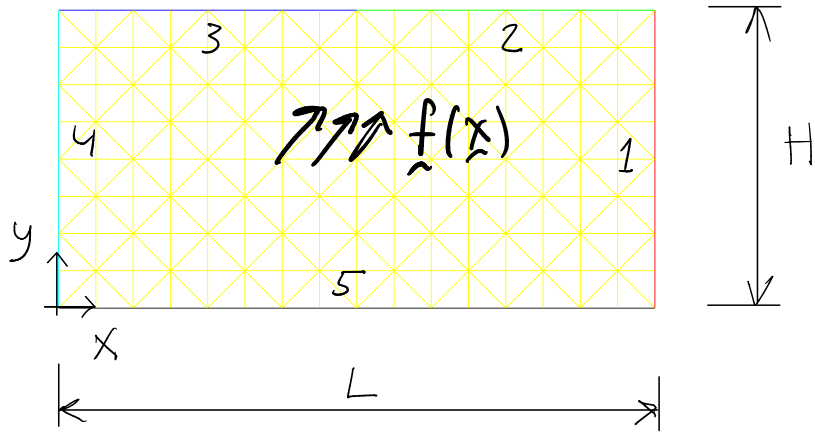

Figure 1: Illustration of a discretized 2D problem domain with boundary segments 1-5. In this task, we consider plane strain elasticity with body force $\boldsymbol{f}(\boldsymbol{x})$ derived from the method of manufactured solutions.

Problem formulation

Consider again the rectangular domain from CA1 as illustrated in the Figure 1. The domain is of dimension $L \times H = 40 \times 20$ mm. Loading will be constructed using the method of manufatured solutions, as described in Appendix. Assume plane strain conditions and isotropic linear elastic response. Choose suitable values for the Elastic parameters.

The task is to study accuracy and convergence with mesh-refinement. In particular, you are also to carry out adaptive mesh-refinement using a posteriori error estimation.

In order to accommodate mesh-refinement, the mesh needs to be stored in a suitable format. Without presenting this format in detail, a coarse mesh is supplied in the .mat-file `grid0.mat`. The grid is stored as a structure variable in MATLAB, and you will not need to work in that data structure. Note the numbering of the boundary segments 1-5 as illustrated in the figure.

In addition to the initial mesh (`grid0.mat`), the following Matlab functions are available for downloading:

1. `grid2CALFEM.m` – Converts the grid to CALFEM format
2. `plant6e.m` – A 6-node triangular element
3. `refinegrid.m` – Refines selected elements of the grid
4. `lin2quad.m` – Projects a linear approximation onto a quadratic topology

Specific tasks

1. Use the method of manufacured solutions described in Appendix, and construct loading data for which you know the exact solution $\boldsymbol{u}$. In particular, determine the energy norm $\|\boldsymbol{u}\|_a$.

   Please be inspired by the example, but construct your own manufacured solution.

2. Study the convergence of the relative error $\|\boldsymbol{e}\|_a/\|\boldsymbol{u}\|_a$ for a sequence of meshes. Try both with linear and quadratic element approximations. Plot the relative error vs. number of degrees of freedom, N_{dof}, in a log-log diagram. Estimate the rate of convergence q (in $\|\boldsymbol{e}\|_a \le Ch^q$).

   Hint 1: If you construct the problem such that $\boldsymbol{u}^{\text{p}} = \boldsymbol{0}$, you are able to express $\|\boldsymbol{e}\|_a$ in terms of $\|\boldsymbol{u}_h\|_a$ and $\|\boldsymbol{u}\|_a$. Doning so, it is sufficient to store the scalar value $\|\boldsymbol{u}\|_a$ and there is no need to compare $\boldsymbol{u}$ and $\boldsymbol{u}_h$ pointwise. $\square$

   Hint 2: By assuming a uniform mesh we can approximate $h \approx \sqrt{2LH/N_{\text{dof}}}$. Thereby, in the presented log-log diagram we obtain the slope $-q/2$, since

   $$\|\boldsymbol{e}\|_a \approx Ch^q \approx \tilde{C} N_{\text{dof}}^{-q/2} \quad \Rightarrow \log \|\boldsymbol{e}\|_a \approx \log \tilde{C} - \frac{q}{2} \log N_{\text{dof}},$$

   with the new constant $\tilde{C} = C\,(2LH)^{q/2}$. $\square$

3. Compute an estimate of the error in energy norm for linear elements by comparing it with quadratic elements. Plot the effectivity index

   $$\eta = \frac{\text{estimated error}}{\text{exact error}}$$

   vs. number of degrees of freedom for a sequence of meshes based on linear elements. The exact error is to be computed based on the overkill solution.

   Hint 3: The nodal values of the approximate error can be computed as

   $$\underline{\widetilde{\boldsymbol{e}}} = \underline{\boldsymbol{a}}_{\text{quad}} - \underline{\boldsymbol{a}}_{\text{lin}}$$

   where $\underline{\boldsymbol{a}}_{\text{quad}}$ is a solution on the quadratic topology and $\underline{\boldsymbol{a}}_{\text{lin}}$ is the linear approximation projected onto the quadratic topology, cf. `lin2quad.m`.

4. Compute the local error contributions and construct an adaptive algorithm refining the elements with the largest contributions recursively, until the estimated error in energy norm is below a required tolerance. Compare the convergence of the error to that of uniform refinement.

   Hint 4: The approximate error can be evaluated in terms of element contributions as

   $$\|\boldsymbol{e}\|_{a,e} \approx \sqrt{\underline{\widetilde{\boldsymbol{e}}}^{eT}\, \widetilde{\boldsymbol{K}}^e\, \underline{\widetilde{\boldsymbol{e}}}^e}$$

   thus defining suitable error indicators.

Appendix: Method of Manufactured Solutions

For the purpose of code verification, or in order to asses the exact error for a set of numerical approximations, it is useful to consider a problem where the exact solution is known. One way to construct such a problem is to use the method of Manufactured sol, which will be described briefly below.

We consider linear isotropic elasticity on domain Ω with boundary $\partial\Omega = \Gamma_{\mathrm{D}} \bigcup \Gamma_{\mathrm{N}}$,

$$-\boldsymbol{\sigma}\left(\boldsymbol{\epsilon}[\boldsymbol{u}]\right) \cdot \boldsymbol{\nabla} = \boldsymbol{f} \quad \text{in } \Omega, \tag{1}$$

$$\boldsymbol{\sigma}\left(\boldsymbol{\epsilon}[\boldsymbol{u}]\right) \cdot \boldsymbol{n} = \boldsymbol{t}^{\mathrm{p}} \quad \text{on } \Gamma_{\mathrm{N}}, \tag{2}$$

$$\boldsymbol{u} = \boldsymbol{u}^{\mathrm{p}} \quad \text{on } \Gamma_{\mathrm{D}}. \tag{3}$$

Furhermore, we assume isotropic response

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = 2\mu\boldsymbol{\epsilon} + \lambda\left(\boldsymbol{I} : \boldsymbol{\epsilon}\right)\boldsymbol{I} \tag{4}$$

with uniform Lamé parameters μ and λ.

For this chosen problem, our goal is now to construct the problem by determining a load case; $\boldsymbol{f}$, $\boldsymbol{t}^{\mathrm{p}}$(and Γ_{N}) and $\boldsymbol{u}^{\mathrm{p}}$(and Γ_{D}), for which we know the exact solution.

The method of munufactured solution is now adopted as follows:

1. *Select* An analytical solution $\boldsymbol{u}^{\mathrm{m}}(\boldsymbol{x})$ being twice differentiable and not coinciding with the finite element discretizations.

2. *Choose* Γ_{D} and *set* the boundary data in Eq. (3) as

$$\boldsymbol{u}^{\mathrm{p}}(\boldsymbol{x}) = \boldsymbol{u}^{\mathrm{m}}.$$

3. *Calculate* analytically the body force in Eq.(1) as

$$\boldsymbol{f}(\boldsymbol{x}) = -\boldsymbol{\sigma}(\boldsymbol{\epsilon}[\boldsymbol{u}^{\mathrm{m}}]) \cdot \boldsymbol{\nabla}.$$

4. If there is a Γ_{D}, calculate analytically the prescribed traction in Eq.(2) as

$$\boldsymbol{t}^{\mathrm{p}}(\boldsymbol{x}) = \boldsymbol{\sigma}(\boldsymbol{\epsilon}[\boldsymbol{u}^{\mathrm{m}}]) \cdot \boldsymbol{n}$$

Constructing the data to the problem ($\boldsymbol{f}$,$\boldsymbol{t}^{\mathrm{p}}$,$\boldsymbol{u}^{\mathrm{p}}$) this way guarantees that $\boldsymbol{u}^{\mathrm{m}}$ satisfies the strong form of the problem. Hence $\boldsymbol{u} = \boldsymbol{u}^{\mathrm{m}}$ is the exact solution to Eqs. (1-4).

Example in 2D

We shall now give a simple example of a manufactured solution for a plane strain problem on the rectangular domain $(x, y) \in [0, L] \times [0, H]$.

1. We choose the following manufactures solution:

$$u_x^{\mathrm{m}} = \sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right), \quad u_y = 0.$$

2. Choosing all boundaries to be of Dirichlet type, we identify

$$\boldsymbol{u}^{\mathrm{p}} = \boldsymbol{0}$$

on $\Gamma_{\mathrm{D}} = \partial\Omega$ from the choice of $\boldsymbol{u}^{\mathrm{m}}$ that vanishes whenever $x = 0$, $x = L$, $y = 0$ or $y = H$.

3. Differentiation gives

$$\frac{\partial u_x^{\rm m}}{\partial x} = \frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right) \qquad \frac{\partial u_x^{\rm m}}{\partial y} = \frac{\pi}{H}\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi y}{H}\right), \tag{5}$$

$$\frac{\partial u_y^{\rm m}}{\partial x} = 0 \qquad \frac{\partial u_y^{\rm m}}{\partial y} = 0, \tag{6}$$

strains

$$\epsilon_{xx}^{\rm m} = \frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right) \qquad \epsilon_{xy}^{\rm m} = \epsilon_{yx}^{\rm m} = \frac{\pi}{2H}\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi y}{H}\right), \tag{7}$$

$$\epsilon_{yy}^{\rm m} = 0 \qquad \epsilon_{\rm vol}^{\rm m} = \frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right), \tag{8}$$

stresses

$$\sigma_{xx}^{\rm m} = \Big(2\mu+\lambda\Big)\frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right) \quad \sigma_{xy}^{\rm m} = \sigma_{yx}^{\rm m} = \mu\frac{\pi}{H}\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi y}{H}\right), \tag{9}$$

$$\sigma_{yy}^{\rm m} = \lambda\frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right), \tag{10}$$

and, finally, body force components

$$f_x(x,y) = -\frac{\partial \sigma_{xx}^{\rm m}}{\partial x} - \frac{\partial \sigma_{xy}^{\rm m}}{\partial y} = \pi^2\left(\frac{2\mu+\lambda}{L^2} + \frac{\mu}{H^2}\right)\sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi y}{H}\right) \tag{11}$$

$$f_y(x,y) = -\frac{\partial \sigma_{yx}^{\rm m}}{\partial x} - \frac{\partial \sigma_{yy}^{\rm m}}{\partial y} = -\frac{\pi^2(\mu+\lambda)}{HL}\cos\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi y}{H}\right) \tag{12}$$

4. Finally, since all boundaries are of Dirichlet type, there is no need to calculate prescribed traction.